

Week 5 – Regression Practical

What I did

- I tried to predict the next week's quiz score.
- I used 8 features from the dataset.
- Missing values were filled using the median.
- I used 75% of the data for training and 25% for testing.
- I tested DummyRegressor, Linear Regression and Ridge.

Results

Model	MAE	RMSE	R ²
DummyRegressor	11.62	13.72	-0.10
Linear Regression	6.81	8.29	0.60
Ridge	6.81	8.29	0.60

What I found

- Linear Regression and Ridge were better than the baseline.
- Both had an MAE of 6.81.
- The actual vs predicted plot showed a clear general pattern.
- The residual plot also showed some larger errors.

Different train/test splits

I tested the model with different random splits.

- MAE was between 6.81 and 7.56.
- R² was between 0.60 and 0.69.
- The results stayed quite similar.

Conclusion

Linear Regression worked better than the baseline. Ridge gave almost the same result. The model can predict the next quiz score reasonably, but the predictions are not exact.